

ITA06 - Machine Learning

Exp :14. To predict house prices using the Linear Regression algorithm.

Procedure

1. Import the Linear Regression model.
2. Create the dataset and target values.
3. Train the model using the training data.
4. Predict the house price for a new input.
5. Display the predicted result.

Dataset 1: House Size, Bedrooms, and Age

House Size (sq.ft) Bedrooms House Age (Years) House Price (₹ Lakhs)

800	2	15	35
1000	2	12	42
1200	3	10	52
1400	3	8	63
1600	4	6	75
1800	4	5	88
2000	5	4	102
2200	5	3	118
2400	6	2	135
2600	6	1	150

```

[1]
✓ 1s
from sklearn.linear_model import LinearRegression
import numpy as np

X = np.array([
    [800,2,15],
    [1000,2,12],
    [1200,3,10],
    [1400,3,8],
    [1600,4,6],
    [1800,4,5],
    [2000,5,4],
    [2200,5,3],
    [2400,6,2],
    [2600,6,1]
])

y = np.array([35,42,52,63,75,88,102,118,135,150])

model = LinearRegression()
model.fit(X, y)

new_sample = np.array([[1700,4,5]])
prediction = model.predict(new_sample)

print("Predicted House Price:", prediction[0], "Lakhs")

```

Predicted House Price: 78.84528301886792 Lakhs

Dataset 2: House Area, Distance from City, and Parking

Area (sq.ft) Distance from City (km) Parking Spaces House Price (₹ Lakhs)

900	20	1	40
1100	18	1	48
1300	16	2	60
1500	14	2	74
1700	12	2	88
1900	10	3	104
2100	8	3	122
2300	6	4	142
2500	5	4	165
2700	3	5	190

```
1 from sklearn.linear_model import LinearRegression
0s import numpy as np

X = np.array([
    [900,20,1],
    [1100,18,1],
    [1300,16,2],
    [1500,14,2],
    [1700,12,2],
    [1900,10,3],
    [2100,8,3],
    [2300,6,4],
    [2500,5,4],
    [2700,3,5]
])

y = np.array([40,48,60,74,88,104,122,142,165,190])

model = LinearRegression()
model.fit(X, y)

new_sample = np.array([[1800,11,3]])
prediction = model.predict(new_sample)

print("Predicted House Price:", prediction[0], "Lakhs")

... Predicted House Price: 102.33333333333337 Lakhs
```

Dataset 3: House Area, Floors, and Garden Size

House Area (sq.ft) Floors Garden Area (sq.ft) House Price (₹ Lakhs)

1000	1	100	45
1200	1	120	55
1400	2	150	68
1600	2	180	82
1800	2	200	96
2000	3	240	112
2200	3	280	130
2400	3	320	150
2600	4	350	172
2800	4	400	196

```

js ▶ from sklearn.linear_model import LinearRegression
import numpy as np

x = np.array([
    [1000,1,100],
    [1200,1,120],
    [1400,2,150],
    [1600,2,180],
    [1800,2,200],
    [2000,3,240],
    [2200,3,280],
    [2400,3,320],
    [2600,4,350],
    [2800,4,400]
])

y = np.array([45,55,68,82,96,112,130,150,172,196])

model = LinearRegression()
model.fit(X, y)

new_sample = np.array([[2100,3,250]])
prediction = model.predict(new_sample)

print("Predicted House Price:", prediction[0], "Lakhs")
... Predicted House Price: 117.79090563165914 Lakhs

```

Dataset 4: Apartment Price Prediction

Apartment Size (sq.ft) Floor Number Building Age (Years) House Price (₹ Lakhs)

700	1	15	30
850	2	12	38
1000	3	10	48
1150	4	8	60
1300	5	7	72
1450	6	5	86
1600	7	4	100
1750	8	3	116
1900	9	2	134
2050	10	1	154

```
4] 0s from sklearn.linear_model import LinearRegression
import numpy as np

x = np.array([
    [700,1,15],
    [850,2,12],
    [1000,3,10],
    [1150,4,8],
    [1300,5,7],
    [1450,6,5],
    [1600,7,4],
    [1750,8,3],
    [1900,9,2],
    [2050,10,1]
])

y = np.array([30,38,48,60,72,86,100,116,134,154])

model = LinearRegression()
model.fit(X, y)

new_sample = np.array([[1500,6,4]])
prediction = model.predict(new_sample)

print("Predicted Apartment Price:", prediction[0], "Lakhs")
```

✓ ... Predicted Apartment Price: 87.030923695058 Lakhs

Dataset 5: Independent House Price Prediction

Plot Area (sq.ft) Built-up Area (sq.ft) Number of Bathrooms House Price (₹ Lakhs)

1200	900	2	48
1400	1100	2	60
1600	1300	3	74
1800	1500	3	90
2000	1700	4	108
2200	1900	4	128
2400	2100	5	150
2600	2300	5	174
2800	2500	6	200
3000	2700	6	228

```
5] from sklearn.linear_model import LinearRegression
0s import numpy as np

X = np.array([
    [1200,900,2],
    [1400,1100,2],
    [1600,1300,3],
    [1800,1500,3],
    [2000,1700,4],
    [2200,1900,4],
    [2400,2100,5],
    [2600,2300,5],
    [2800,2500,6],
    [3000,2700,6]
])

y = np.array([48,60,74,90,108,128,150,174,200,228])

model = LinearRegression()
model.fit(X, y)

new_sample = np.array([[2300,2000,4]])
prediction = model.predict(new_sample)

print("Predicted House Price:", prediction[0], "Lakhs")
```

✓ ... Predicted House Price: 146.0 Lakhs